

1. Write a program that asks the user for their age and checks if they are eligible to vote (18 years and older). Print a message based on their eligibility.

ANS

```
age = int(input("Enter age : "))

if age >= 18:
    print("Eligible to Vote")
else:
    print("Not eligible to Vote")
```

OUTPUT

```
Enter age : 23
Eligible to Vote
```

2. Write a program that takes a year as input and checks if it is a leap year or not. Print the result.

ANS

```
year = int(input("Enter year : "))

if year % 4 == 0:
    print("The year" , year , "year is a leap year")
else:
    print("The year", year, "year is not a leap year")
```

OUTPUT

```
Enter year : 2026
The year 2026 year is not a leap year
```

3. Write a program that takes a number as input and checks if it is divisible by 5. Print a message if it is divisible or not.

ANS

```
n = int(input("enter a number: "))

if n % 5 == 0:
    print("divisible by 5")
else:
    print("not divisible by 5")
```

OUTPUT

```
enter a number: 25
divisible by 5
```

4. Write a program that takes a character as input and checks if it is a vowel or consonant. Print the result.

ANS

```
ch = input("enter a character: ")

if ch in 'aeiouAEIOU':
    print("it is a vowel")
else:
    print("it is a consonent")
```

OUTPUT

```
enter a character: E
```

it is a vowel

5. Write a program that takes a password as input and checks if it is strong (at least 8 characters). Print a message based on the result.

ANS

```
password = input("enter a password:")

if len(password) >= 8:
    print("strong password")
else:
    print("weak password it should be at least 8 characters")
```

OUTPUT

```
enter a password:124@#$SDFD@31
strong password
```

6. Write a program that takes a grade (A, B, C, D, F) as input and uses match-case to print a message corresponding to the grade (e.g., "Excellent" for A, "Good" for B, etc.).

ANS

```
grade = input("enter your grade(A/B/C/F):").upper()

match grade:
    case 'A':
        print("Excellent")
    case 'B':
        print("Good")
    case 'C':
        print("Average")
    case 'F':
        print("Fail")
    case _:
        print("Invalid grade")
```

OUTPUT

```
enter your grade(A/B/C/F):A
Excellent
```

7. Write a program that takes a traffic light color (Red, Yellow, Green) as input and uses match-case to print the corresponding action (e.g., "Stop" for Red, "Wait" for Yellow, etc.).

ANS

```
colors = input("Enter traffic light color:").capitalize()

match colors:
    case "Red":
        print("Stop")
    case "Yellow":
        print("Wait")
    case "Green":
        print("Go")
    case _:
        print("Invalid color")
```

OUTPUT

Enter traffic light color:Green
Go

8. Write a program that takes a number as input and uses a for loop to calculate its factorial. Print the result.

ANS

```
n=int(input("Enter a number:"))
fact = 1

for i in range(1,n+1):
    fact *= i

print("Factorial =",fact)
```

OUTPUT

Enter a number:8
Factorial = 40320

9. Write a program that uses a for loop to print all even numbers between 1 and 20.

ANS

```
for n in range(1,21):
    if n %2 == 0:
        print(n)
```

OUTPUT

2
4
6
8
10
12
14
16
18
20

10. Write a program that uses a while loop to create a countdown timer from 10 to 0.

Print each number.

ANS

```
count = 10
while count >=0:
    print(count)
    count -= 1
```

OUTPUT

10
9
8
7
6
5
4
3

2
1
0

11. Write a program that uses a for loop to count the number of vowels in a given string. Print the count.

ANS

```
text = input("Enter a string: ")  
count = 0
```

```
for ch in text:  
    if ch in 'aeiouAEIOU':  
        count += 1
```

```
print("Number of vowels:", count)
```

OUTPUT

```
Enter a string: TRISHA  
Number of vowels: 2
```

12. Write a program that uses a for loop to print numbers from 1 to 10, but skips printing the number 5 (use continue) and stops the loop if the number 8 is reached (use break).

ANS

```
for n in range(1,11):  
    if n == 5:  
        continue  
    if n == 8:  
        break  
    print(n)
```

OUTPUT

1
2
3
4
6
7